

Mahmoud Hisham Almodalal

Backend & AI Engineer

Computer Engineer focused on backend systems, scalable SaaS architecture, and AI-powered applications. My work spans multi-tenant Django platforms, PostgreSQL data modeling, production deployment, offline-capable web applications, and RAG/LLM support systems.

CORE EXPERTISE

Backend: Python, Django, DRF, FastAPI, PHP, Laravel, REST APIs, WebSockets
Databases: PostgreSQL, MySQL, SQLite, Oracle DB, Redis
AI: LLMs, RAG, OpenAI, Gemini, DeepSeek, YOLOv8, DeepSort, InsightFace
DevOps: Docker, GitHub Actions CI/CD, Nginx, Linux, Cloudflare, Git/GitHub
Frontend: React, Next.js, TypeScript, JavaScript, HTML5, CSS3, Bootstrap

SELECTED WORK

EduTracker	Multi-tenant bilingual school management SaaS Django / PostgreSQL / React / Redis / Docker
Noor Al-Huda	Offline-capable Quran center management for 300+ students DRF / PostgreSQL / Next.js / Dexie.js
clientMORE	Omnichannel bilingual AI customer-support SaaS FastAPI / RAG / LLMs / WebSockets / ChromaDB

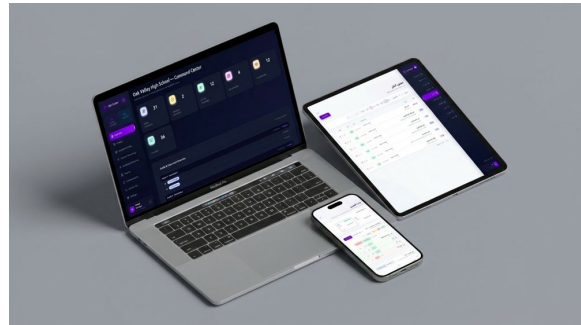
CONTACT & LINKS

ghg17273@gmail.com | [+972 597 747 084](tel:+972597747084) | [GitHub](#)

Education: B.Sc. in Computer Engineering, Islamic University of Gaza - March 2026 - GPA 91%

EduTracker

All-in-one bilingual school management SaaS



EduTracker centralizes administration, tuition tracking, communication, and role-specific dashboards for regional managers, administrators, teachers, students, and parents. The platform includes offline support and automatic sync.

ENGINEERING HIGHLIGHTS

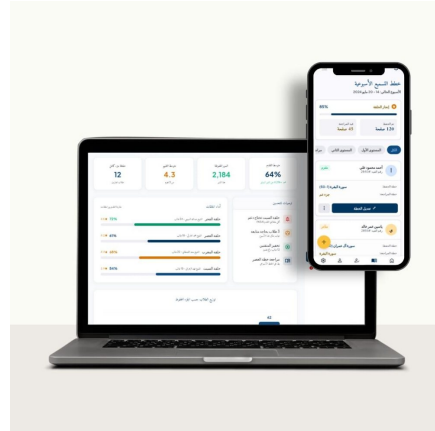
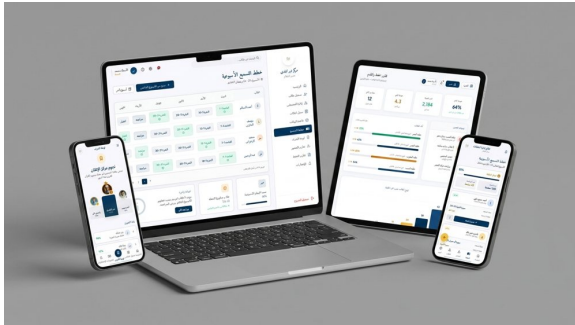
- Clean Architecture backend following the HackSoft Django Styleguide with Service / Selector / Policy layers.
- 32-entity PostgreSQL schema with Arabic full-text search and multi-role RBAC for managers, teachers, secretaries, and guardians.
- Celery and Redis for background processing, plus full CI/CD using GitHub Actions, Docker Compose, Nginx, SSL, and Cloudflare.
- React 19 + TypeScript frontend paired with Django REST Framework APIs.

Stack: Django, DRF, PostgreSQL, React 19, Vite, Tailwind CSS, Celery, Redis, Docker, Nginx, Cloudflare, GitHub Actions

[Open live platform](#) | [EduTracker LinkedIn](#) | Repository: Private

Noor Al-Huda

Quran-memorization center management platform



A digital management workflow serving 300+ students and replacing paper logs. The platform is designed to remain usable offline and synchronize data when connectivity becomes available.

PRODUCT & TECHNICAL HIGHLIGHTS

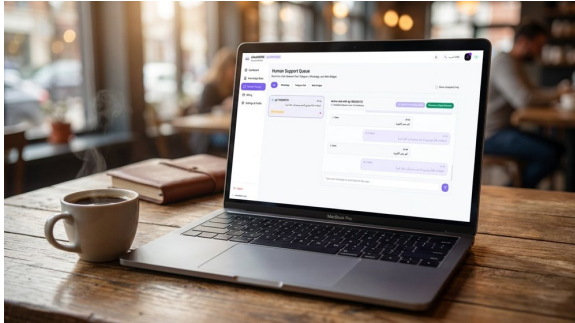
- Tracks memorization progress and academic activity in a structured workflow.
- Generates automated PDF report cards for students.
- Supports targeted announcements for parents, students, and teachers.
- Uses Dexie.js for offline-capable client storage with a Django REST Framework and PostgreSQL backend.

Stack: Next.js, React, TypeScript, Tailwind CSS, Dexie.js, Django REST Framework, PostgreSQL, Docker, Render, Vercel

[Open live platform](#) | [View GitHub repository](#)

clientMORE

Omnichannel bilingual AI customer-support SaaS



clientMORE automates Arabic/English business communication using a document-driven assistant with smart escalation to live agents. It combines multiple messaging channels with a Hybrid RAG pipeline and multi-provider LLM chain.

AI & SYSTEM HIGHLIGHTS

- Unified messaging router using WebSockets, Telegram, and WhatsApp integrations.
- Hybrid RAG pipeline backed by ChromaDB for document-grounded responses.
- Multi-provider LLM chain using Gemini, DeepSeek, and OpenAI.
- Bilingual intent-routing state machine with escalation from AI assistance to live agents.

Stack: Next.js, TypeScript, Tailwind CSS, FastAPI, SQLite, ChromaDB, WebSockets, Telegram API, Twilio, Docker

[Open live platform](#) | [View GitHub repository](#)

PROFESSIONAL BACKGROUND

Experience & Credentials

EXPERIENCE

Computer Engineering Instructor - MET Academy

May 2025 - Present

Teach Calculus, Logic Design, and Computer Architecture to computer science students.

Co-Founder & Backend Developer - EduTracker

Graduation Project

Designed and deployed a multi-tenant school-management SaaS with Django, PostgreSQL, RBAC, CI/CD, and Docker.

Web Development Intern - Ministry of Health, IT Department

Aug 2025 - Mar 2026

Worked across responsive frontend development, PHP/Laravel backend logic, MySQL, and Oracle DB.

AI Scripting Specialist - PDX Ai Scripting

Aug 2023 - Oct 2023

Developed face detection / recognition workflows and optimized AI-processing performance.

Python Intern (ML & AI) - Al-Raja International Training Company

Jul 2022 - Oct 2022

Completed Python, machine learning, AI, algorithm, and supervised software-project training.

Experience verification: [Al-Raja Experience Certificate](#)

CERTIFICATIONS & VERIFICATION

- Machine Learning - HOP International Center, Jordan - [Verify](#)
- Neural Networks and Deep Learning - DeepLearning.AI / Coursera - [Verify](#)
- ICDL - Othman Center, Gaza - [Verify](#)
- 21st Century Skills & Freelancing - IESCO - [Verify](#)
- Django Application Development with SQL - IBM / Coursera - [Verify](#)
- Getting Started with Git and GitHub - IBM / Coursera - [Verify](#)
- Introduction to Software Engineering - IBM / Coursera - [Verify](#)
- English Placement Test (Score: 9/12) - Islamic University of Gaza - [Verify](#)

LEADERSHIP & REFERENCES

- Knowledge Sharing: Organized and led study sessions for peers in Computer Engineering.
- Technical Tutoring: Simplified Data Structures, Algorithms, and Networking topics.
- Reach Education Fund Scholar: Recognized for professionalism and commitment to community service.

Walid Mizyed - Founder & Executive Director, Reach Education Fund - [Recommendation Letter](#)

Dr. Ahmed Mahdi - Assistant Professor, Computer Engineering Dept. (IUG)

Contact: ghg17273@gmail.com | [GitHub](#)